Lecture 4: Hypothesis testing: Basics and P-values

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Suggested Reading: Hogg et al. 2018, Chapter 8.1 to 8.4.

1 Testing a Hypothesis

Introduction: In practice, scientists and statisticians always need to give a true / false answer to an assumption (hypothesis) about the population parameters after collecting some data from that population. This lecture introduces how to use data to verify the hypothesis in a scientific manner so that statisticians are **not likely to** make wrong conclusions. In this section we first introduce some basic definitions and notations.

Definition 1.1 (**Hypothesis, Null hypothesis, Alternative hypothesis**). *In statistics, a hypothesis is a statement about the population distribution.*

Especially, when testing a parameter θ, define the null hypothesis H_0 as $\theta \in A_0$ for some given set A_0 and the alternative hypothesis H_1 as $\theta \in A_1$ for some given set A_1. Here $A_0 \cap A_1 = \emptyset$.

A statistical hypothesis test, called a test for short, uses data to summarize the evidence about a hypothesis, and it will either **accept** the null hypothesis or **reject** the null hypothesis (thus accept the alternative hypothesis) in the end.

Remark

1. A hypothesis should only have true and false answer, and statisticians **must** pick one.
2. We assume that $\theta \in A_0 \cup A_1$ a priori, and $A_0 \cup A_1$ **need not** to be the whole space.
3. The hypothesis are formulated before analyzing the data.
4. The hypothesis should be for population parameter, not the estimator.
5. Before performing test, whether $\mu \in A_0$ or $\mu \in A_1$ is unknown.

Example 1 Testing the mean Suppose the data $X_1, X_2, \cdots, X_n$ are i.i.d. random variables satisfying normal distribution $N(\mu, \sigma^2)$, here μ, σ are unknown. Then the hypothesis can be

1. **One side test**: $H_0 : \mu \leq 0$ and $H_1 : \mu > 0$
2. **Gap one side test**: $H_0 : \mu \leq 0$ and $H_1 : \mu > 3$
3. **Two side test**: $H_0 : \mu = 0$ and $H_1 : \mu \neq 0$
4. **Two point test**: $H_0 : \mu = 0$ and $H_1 : \mu = 1$. (In that case, μ has to be 0 or 1 a priori.)

Remark In practice, we always consider H_0 as the 'normal / trivial / default situation', and H_1 as the abnormal situation (meaning some bad event happens / new scientifical findings). A hypothesis test analyzes whether the data contradict H_0 and suggest that H_1 is true. H_0 is presumed to be true by

default. Under this presumption, if the observed data would be very *unusual*, the evidence supports H_1 and the test will reject H_0 accordingly.

From the above example 1, we can see that the challenge of conducting the test is the unknown μ. Since μ is unknown, we have to estimate μ (e.g., use the sample mean $\overline{X}$). However since $\overline{X}$ is a random variable, μ satisfies H_0 does not mean that $\overline{X}$ satisfies H_0. Therefore, we have to study how to use $\overline{X}$ to draw conclusion about μ.

Definition 1.2 (**Test Statistics**). *Suppose the hypothesis is about a population parameter θ, then a test statistics should be an estimator $\widehat{\theta}$ of θ that summarizes how far that point estimate falls from the parameter value/region in H_0.*

As an example, suppose the following test:

Data $X_1, \cdots, X_n$ are independent normal $N(\mu, \sigma^2)$ random variables, μ unknown and σ^2 is known. We want to test

$$H_0 : \mu = 0 \text{ versus } \ H_1 : \mu \neq 0$$

A good estimator is $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, because we know that, under the null, i.e., when $\mu = 0$,

$$\overline{X} \in \left[-\frac{\sigma}{\sqrt{n}} \times z_{0.025}, \ \frac{\sigma}{\sqrt{n}} \times z_{0.025} \right] \tag{1}$$

with probability 95%, where $z_{0.025}$ is the upper 0.025 quantile of the standard normal distribution. Here $\overline{X}$ is a good test statistics since we can summarize where the test statistics should locate under the null hypothesis. In practice, if the sample mean $\bar{X}$ is far away from the interval $\left[-\frac{\sigma}{\sqrt{n}} \times z_{0.025}, \ \frac{\sigma}{\sqrt{n}} \times z_{0.025} \right]$ (i.e., when $|\bar{X}| > \frac{\sigma}{\sqrt{n}} \times z_{0.025}$), this indicates the null hypothesis is likely to be false, and thus we may choose to reject H_0. Thus we define the rejection region below.

Definition 1.3 (**Rejection Region**). *The test will reject the null hypothesis H_0 if the test statistics $\widehat{\theta} \in R$, and R is called the rejection region.*

The rejection region is typically selected by simultaneously controlling the two types of errors:

Definition 1.4 (**Type-I, Type-II errors, and Power**). *1. The probability of rejecting the null hypothesis H_0 when it is true is defined as the Type-I error, usually denoted as α.*

2. The probability of failing to reject the null hypothesis H_0 when it is false is defined as the Type-II error, usually denoted as β.

3. We call $1-$ Type–II error "power".

The statistician usually controls type-I error probability when she selects the rejection region – setting type-I error probability at a desired level called *significance level*.

Given a rejection region and test statistics, we can perform the hypothesis test following this procedure: (i) first calculate the test statistics from data; (ii) then compare the test statistics with the rejection region, and (iii) reject H_0 if $\widehat{\theta} \in R$.

Continue the above example of testing the population mean $\mu = 0$, if we select the rejection region

$$R = (-\infty, -\frac{\sigma}{\sqrt{n}} \times z_{0.025}] \cup [\frac{\sigma}{\sqrt{n}} \times z_{0.025}, \infty)$$

and reject the test if $\overline{X} \in R$, then we have the Type-I error equals to α.

Notably, as will be demonstrated later, there is always a trade–off between Type–I and Type–II errors. That is, a small Type–I error always results in a small rejection region, thue increase the Type–II error. In practice, we always bound the Type–I error to be inside a tolerance level, like 5%, and try to control type–II error then.

2 Examples of Common Tests

2.1 Test for Mean

We first study the hypothesis test for the normal mean.

Known Variance Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known. We consider the null hypothesis $H_0 : \mu = \mu_0$ and three types of alternative hypotheses below. We use $Z_0 := \frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma}$ as the test statistic. Note that

$$Z_0 = \frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} + \frac{\sqrt{n}(\mu - \mu_0)}{\sigma} = N(0,1) + \text{bias}$$

Note that $Z_0 \sim N(0,1)$ when H_0 is true. Set the significance level as α.

1. $H_1 : \mu \neq \mu_0$. We choose the rejection region

   $$R = (-\infty, -z_{\alpha/2}] \cup [z_{\alpha/2}, \infty)$$

   here $z_{\alpha/2}$ is the upper $\alpha/2$ quantile of $N(0,1)$ distribution. Under H_0,

   $$P\left(Z_0 \in R\right) = P(Z_0 \leq -z_{\alpha/2}) + P(Z_0 \geq z_{\alpha/2}) = \alpha$$

   So the rejection region has significance level α (Type-I error $=\alpha$).

   Under H_1, $|\frac{\sqrt{n}(\mu-\mu_0)}{\sigma}| \to \infty$, so it is likely that $Z_0 \geq z_{\alpha/2}$ (if $\mu > \mu_0$) or $Z_0 \leq -z_{\alpha/2}$ (if $\mu < \mu_0$).

2. $H_1 : \mu > \mu_0$. In such case we have the rejection region

$$R = [z_\alpha, \infty)$$

and reject the test if $Z_0 \in R$. Under H_0,

$$P(Z_0 \in R) = P(Z_0 \geq z_\alpha) = \alpha$$

So the rejection region has significance level α.

Under H_1, $\mu > \mu_0$, so the bias $\frac{\sqrt{n}(\mu-\mu_0)}{\sigma} \to \infty$, and it is likely for $Z_0 \geq z_\alpha$.

3. $H_1 : \mu < \mu_0$. In that case we choose the rejection region

$$R = (-\infty, -z_\alpha]$$

and reject the test if $Z_0 \in R$. Under H_0,

$$P(Z_0 \in R) = P(Z_0 \leq -z_\alpha) = \alpha$$

Under H_1, $\mu < \mu_0$, so the bias $\frac{\sqrt{n}(\mu-\mu_0)}{\sigma} \to -\infty$, and it is likely for $Z_0 \leq -z_\alpha$.

Unknown Variance Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with σ^2 unknown. We consider the null hypothesis $H_0 : \mu = \mu_0$ and three types of alternative hypotheses. We use $T_0 := \sqrt{n}\frac{\bar{X}-\mu_0}{S}$ with S^2 being the sample variance as the test statistic. Still,

$$T_0 = \frac{\sqrt{n}(\overline{X} - \mu)}{S} + \frac{\sqrt{n}(\mu - \mu_0)}{S}$$

So $T_0 \sim t(n-1)$ when H_0 is true.

1. $H_1 : \mu \neq \mu_0$. Choose the rejection region

$$R = (-\infty, -t_{\alpha/2,n-1}] \cup [t_{\alpha/2,n-1}, \infty)$$

here $t_{\alpha/2,n-1}$ is the upper $\alpha/2$ quantile of the t_{n-1} distribution. Under H_0,

$$P(T_0 \in R) = P(T_0 \leq -t_{\alpha/2,n-1}) + P(T_0 \geq t_{\alpha/2,n-1}) = \alpha$$

Under H_1, $|\frac{\sqrt{n}(\mu-\mu_0)}{S}| \to \infty$, so it is likely for T_0 to be in R.

2. $H_1 : \mu > \mu_0$. Choose the rejection region

$$R = [t_{\alpha,n-1}, \infty)$$

Under the null, T_0 has t_{n-1} distribution, so

$$P(T_0 \in R) = P(T_0 \geq t_{\alpha,n-1}) = \alpha$$

On the other hand, under H_1, $\frac{\sqrt{n}(\mu-\mu_0)}{S} \to \infty$, so the T_0 is likely to be in R.

3. $H_1 : \mu < \mu_0$. Choose the rejection region

$$R = (-\infty, -t_{\alpha,n-1}]$$

Under the null, T_0 has t_{n-1} distribution, so

$$P\left(T_0 \in R\right) = P(T_0 \leq -t_{\alpha,n-1}) = \alpha$$

On the other hand, under H_1, $\frac{\sqrt{n}(\mu-\mu_0)}{S} \to -\infty$, so it is likely for T_0 to be in R.

When the data distribution is not normal, we use CLT to approximate the distribution of $\bar{X}$ and thus previous tests for mean can still be performed with an approximate Type-I error being α.

Remark In the previous example, we can see that your rejection region depends on H_0 and H_1. Despite you can select your own rejection region, you should assure that:

i. Under H_0, the significance level equals / approximately equals α.

ii. Under H_1, you test statistics should (asymptotically) be in R.

2.2 Test for Proportion

Suppose $X_1, X_2, \cdots, X_n \sim Bernoulli(p)$. We consider the null hypothesis $H_0 : p = p_0 \in (0, 1)$. Recall from central limit theorem that $\frac{\sqrt{n}(\widehat{p}-p)}{\sqrt{p(1-p)}} \to_d N(0,1)$ and $\widehat{p} \to_p p$, thus from Slutsky's theorem:

$$\frac{\sqrt{n}(\widehat{p}-p)}{\sqrt{\widehat{p}(1-\widehat{p})}} \to_d N(0,1), \text{ here } \widehat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

So we can consider the test statistics

$$A_0 = \frac{\sqrt{n}(\widehat{p}-p_0)}{\sqrt{\widehat{p}(1-\widehat{p})}} = \frac{\sqrt{n}(\widehat{p}-p)}{\sqrt{\widehat{p}(1-\widehat{p})}} + \frac{\sqrt{n}(p-p_0)}{\sqrt{\widehat{p}(1-\widehat{p})}}$$

Under the null, $p = p_0$, so $A_0 \to_d N(0,1)$

1. Consider the alternative H_1: $p \neq p_0$. Define the rejection region

$$R = (-\infty, -z_{\alpha/2}] \cup [z_{\alpha/2}, \infty)$$

with z_α the upper α quantile for standard normal distribution. Then under H_0

$$\lim_{n\to\infty} P(A_0 \in R) = P(Z \in R) = \alpha, \text{ here } Z \sim N(0,1)$$

Under H_1, we have $|\frac{\sqrt{n}(p-p_0)}{\sqrt{\widehat{p}(1-\widehat{p})}}| \to \infty$, so it is likely for A_0 to be in R.

2.3 Test for Variance

Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$. We consider the null hypothesis $H_0 : \sigma^2 = \sigma_0^2$ and three types of alternative hypotheses below. We use $\chi_0 := \frac{(n-1)S^2}{\sigma_0^2}$ as the test statistic. Note that $\chi_0 \sim \chi^2(n-1)$ when H_0 is true. Set the significance level as α.

1. $H_1 : \sigma^2 \neq \sigma_0^2$. Set the rejection region $R := \{\chi_0 > \chi^2_{\alpha/2,n-1} \text{ or } \chi_0 < \chi^2_{1-\alpha/2,n-1}\}$ which guarantees that the type-I error is α.

2. $H_1 : \sigma^2 > \sigma_0^2$. Set the rejection region $R := \{\chi_0 > \chi^2_{\alpha,n-1}\}$.

3. $H_1 : \sigma^2 < \sigma_0^2$. Set the rejection region $R := \{\chi_0 < \chi^2_{1-\alpha,n-1}\}$.

3 $P-$value

We start by its definition.

Definition 3.1. *Suppose we have hypothesis H_0 versus H_1 based on the test statistics T and the rejection region $R = R(\alpha)$, i.e., the rejection region depends on the significance level α. For an observed test statistics value T, the $P-$value is defined by*

$$\widehat{p} = \inf\{\alpha \in [0,1] : T \in R(\alpha)\} \tag{2}$$

The aforementioned formula means "the smallest probability of T inside the rejection region." Alternatively, $\widehat{p}$ is considered to be the largest possible α(probability) that T is not inside $R(\alpha)$, i.e., fail to reject H_0.

Example 2 Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known. We consider the null hypothesis $H_0 : \mu = \mu_0$ and three types of alternative hypotheses below. We use $Z_0 := \frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma}$ as the test statistic. Set the significance level as α.

1. $H_1 : \mu \neq \mu_0$. We choose the rejection region

$$R(\alpha) = (-\infty, -z_{\alpha/2}] \cup [z_{\alpha/2}, \infty)$$

here $z_{\alpha/2}$ is the upper $\alpha/2$ quantile of $N(0,1)$ distribution. By definition, for the observed test statistics value Z_0, the $P-$value is

$$\widehat{p} = \inf \{\alpha \in [0,1] : Z_0 \in R(\alpha)\}$$

Define Φ as the cdf of a standard normal random variable.

$$\begin{aligned} Z_0 \in R(\alpha) \Longleftrightarrow |Z_0| \geq z_{\alpha/2} \Longleftrightarrow \Phi(|Z_0|) - \Phi(-|Z_0|) \geq \Phi(z_{\alpha/2}) - \Phi(-z_{\alpha/2}) = 1 - \alpha \\ \Rightarrow \alpha \geq 1 - \Phi(|Z_0|) + \Phi(-|Z_0|) \end{aligned} \tag{3}$$

(think about this in detail after the class). This means that $P-$value is

$$\hat{p} = 1 - \Phi(|Z_0|) + \Phi(-|Z_0|) = 2\Phi(-|Z_0|) = \mathbf{P}_{Z \sim N(0,1)}[|Z| \geq |Z_0|]$$

This means that for two-sided test, the $P-$value is the probability that the distribution of test statistics (under H_0) is more extreme than the observed test statistics value Z_0.

2. $H_1 : \mu > \mu_0$. In such case we have the rejection region

$$R(\alpha) = [z_\alpha, \infty)$$

Therefore, by definition

$$\widehat{p} = \inf\{\alpha \in [0,1] : Z_0 \in R(\alpha)\} = \inf\{\alpha \in [0,1] : Z_0 \geq z_\alpha\} = \inf\{\alpha \in [0,1] : \Phi(Z_0) \geq 1 - \alpha\}$$

so the $P-$value is $\alpha = 1 - \Phi(Z_0)$, i.e., $\mathbf{P}_{Z \sim N(0,1)}[Z \geq Z_0]$.

3. $H_1 : \mu < \mu_0$. In that case we choose the rejection region

$$R(\alpha) = (-\infty, -z_\alpha]$$

Therefore, the $P-$value is

$$\widehat{p} = \inf\{\alpha \in [0,1] : Z_0 \in R(\alpha)\} = \inf\{\alpha \in [0,1] : Z_0 \leq -z_\alpha\} = \inf\{\alpha \in [0,1] : \Phi(Z_0) \leq \alpha\}$$

so the $P-$value is given by $\hat{p} = \Phi(Z_0)$, which is $\mathbf{P}_{Z\sim N(0,1)}[Z \leq Z_0]$.

Example 3 Now suppose $X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2)$ with σ^2 unknown. Define $\widehat{\mu} = \frac{1}{n}\sum_{i=1}^n X_i$ and $S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \overline{X})^2$. Suppose the null hypothesis $H_0 : \mu = \mu_0$. Define the test statistics $\widehat{T} = \frac{\sqrt{n}(\widehat{\mu}-\mu_0)}{S}$. Under H_0, we know that $\widehat{T} \sim t_{n-1}$.

(i). Suppose the alternative $H_1 : \mu \neq \mu_0$, then we define the rejection region $(-\infty, -t_{\alpha/2,n-1}) \cup (t_{\alpha/2,n-1}, \infty)$, then the P-value is given by

$$\begin{aligned}\widehat{p} = \inf\{\alpha : \widehat{T} \in R_\alpha\} = \inf\{\alpha : |\widehat{T}| > t_{\alpha/2}\} = \inf\{\alpha : \Psi(|\widehat{T}|) - \Psi(-|\widehat{T}|) > \Psi(t_{\alpha/2}) - \Psi(-t_{\alpha/2})\} \\ = \inf\{\alpha : \Psi(|\widehat{T}|) - \Psi(-|\widehat{T}|) > 1-\alpha\} \\ \Rightarrow \alpha = 1 - \Psi(|\widehat{T}|) + \Psi(-|\widehat{T}|)\end{aligned}$$

Here $\Psi(x)$ is the cumulative distribution function of a t_{n-1} random variable.

(ii). Suppose the alternative $H_1 : \mu > \mu_0$, then the rejection region is $R_\alpha = (t_{\alpha,n-1}, \infty)$, so the P-value is given by

$$\begin{aligned}\widehat{p} = \inf\{\alpha : \widehat{T} \in R_\alpha\} = \inf\{\alpha : \widehat{T} > t_\alpha\} = \inf\{\alpha : \Psi(\widehat{T}) > \Psi(t_\alpha)(= 1-\alpha)\} \\ \Rightarrow \alpha = 1 - \Psi(\widehat{T})\end{aligned}$$

$P-$value has a special property, i.e., suppose the null hypothesis H_0 holds true. Then the $P-$value $\widehat{p}$ has uniform distribution.

Lemma 3.1 (in the probability review class)**.** *Suppose X is a continuous random variable. Besides, its cumulative distribution function F_X is strictly increasing in $\mathbf{R}$. Then $F_X(X)$ has uniform distribution on $[0,1]$, i.e.,*

$$\mathbf{P}\left(F_X(X) \leq x\right) = \begin{cases} 0, & \text{if } x \leq 0 \\ x, & \text{if } x \in [0,1] \\ 1, & \text{if } x > 1 \end{cases}$$

Theorem 3.2. *Suppose the null hypothesis H_0 holds true and the estimator $\widehat{\theta}$ satisfies the central limit theorem, i.e.,*

$$\frac{\widehat{\theta} - \theta_0}{\widehat{\sigma}} \to_d N(0,1),$$

here θ_0 is the real underlying parameter and $\widehat{\sigma}^2$ is an estimator for $Var(\widehat{\theta})$.

Then for the two side test, we have

$$\widehat{p} \to_d Unif(0,1) \tag{4}$$

Proof Define Z as a standard normal random variable. We know that $1 - \Big(\Phi\big(|\frac{\widehat{\theta}-\theta_0}{\widehat{\sigma}}|\big) - \Phi\big(-|\frac{\widehat{\theta}-\theta_0}{\widehat{\sigma}}|\big)\Big)$

is the p-value (see Example 2). For $f(x) = 1 - (\Phi(|x|) - \Phi(-|x|))$ is a continuous function, from the continuous mapping theorem (see Lecture 1)

$$1 - \left(\Phi\big(|\frac{\widehat{\theta} - \theta_0}{\widehat{\sigma}}|\big) - \Phi\big(- |\frac{\widehat{\theta} - \theta_0}{\widehat{\sigma}}|\big) \right) \to_d 1 - \left(\Phi\left(|Z|\right) - \Phi\left(-|Z|\right) \right)$$

For $|Z|$'s cumulative distribution function is $\Phi(x) - \Phi(-x)$ for $x \geq 0$, from theorem 2.4 in Lecture 1 (the probability review), $1 - \left(\Phi\left(|Z|\right) - \Phi\left(-|Z|\right) \right)$ has uniform distribution on $[0, 1]$. Correspondingly

$$1 - \left(\Phi\big(|\frac{\widehat{\theta} - \theta_0}{\widehat{\sigma}}|\big) - \Phi\big(- |\frac{\widehat{\theta} - \theta_0}{\widehat{\sigma}}|\big) \right) \to_d Unif(0, 1)$$

□

Corollary 3.1 (The alternative situation)**.** *Let us see what happens if the alternative situation happens, i.e., $\theta \neq \theta_0$. In that case*

$$\frac{\widehat{\theta} - \theta_0}{\widehat{\sigma}} = \frac{\widehat{\theta} - \theta}{\widehat{\sigma}} + \frac{\theta - \theta_0}{\widehat{\sigma}}$$

For $\widehat{\sigma} \to_p 0$, we have

$$\left(\Phi\left(|\frac{\widehat{\theta} - \theta_0}{\widehat{\sigma}}| \right) - \Phi\left(-|\frac{\widehat{\theta} - \theta_0}{\widehat{\sigma}}| \right) \right) \to_p \left(\Phi\left(+\infty\right) - \Phi\left(-\infty\right) \right) = 1$$

and the $P-$value tends to 0.

From the corollary, we see that, for one side & two side test, the $P-$value tends to 0 under the alternative situation.

Remark As you can see in the example, the definition of a $P-$value estimator in hypothesis testing is case-by-case, i.e., if the test (especially, rejection region) is different, then you may need to establish a new $P-$value estimator for the test. However, the idea is the same, i.e., under the null hypothesis, the $P-$value should (at least under some situation) have a uniform distribution, and the $P-$value should not be too small with large probability. Under the alternative, the $P-$value should tend to 0.

After acquiring P–value of the test, statisticians can decide accepting or rejecting H_0 via the value of P–value. To see how it works, let us consider the following theorem.

Theorem 3.3. *For a given $\alpha \in (0, 1)$, suppose we reject the null hypothesis when $\widehat{p} < \alpha$. Then*

$$\lim_{n \to \infty} \mathbf{P}\left(\textit{We make a false rejection under } H_0 \right) \leq \alpha$$

Proof For the two side test situation:

$$\lim_{n \to \infty} \mathbf{P}\left(\text{We make a false rejection} \right) = \lim_{n \to \infty} \mathbf{P}\left(\widehat{p} < \alpha \right) = \mathbf{P}(U \leq \alpha) = \alpha, \tag{5}$$

where $U \sim Unif(0,1)$. For the one-side situation

$$\lim_{n\to\infty} \mathbf{P}\left(\text{We make a false rejection}\right) = \lim_{n\to\infty} \mathbf{P}\left(\widehat{p} < \alpha\right) \leq \mathbf{P}(U < \alpha) = \alpha \tag{6}$$

□

From this theorem, by rejecting H_0 when $\widehat{p} < \alpha$, statisticians can control the Type–I error to α.

4 Duality between test and confidence interval*

Corollary 4.1 (Duality between two side test and the confidence interval)**.** *For the two side test, we can solve the inequality, i.e., we do not reject the null if*

$$\widehat{p} \geq \alpha \Longleftrightarrow \frac{|\widehat{\theta} - \theta_0|}{\widehat{\sigma}} \leq z_{\alpha/2} \longrightarrow \theta_0 \in \left[\widehat{\theta} - \widehat{\sigma} z_{\alpha/2},\ \widehat{\theta} + \widehat{\sigma} z_{\alpha/2}\right] \tag{7}$$

here $z_{\alpha/2}$ is the upper $\alpha/2$ quantile of $N(0,1)$. Therefore, we have:

For a two side test, $\widehat{p} \geq \alpha$ is equivalent to: θ_0 is in the $1-\alpha$ confidence interval of θ; and we reject the null hypothesis if $\theta_0 \notin$ the $1-\alpha$ confidence interval of θ.

5 Summary

In summary, for a given hypothesis testing problem, there are usually three (equivalent) ways to perform the test, using the estimator $\hat{\theta}$ which satisfies $\frac{\widehat{\theta}-\theta_0}{\widehat{\sigma}} \to_d N(0,1)$, for two-sided test $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$:

1. Rejection region: $R(\alpha) = (-\infty, \theta_0 - z_{\alpha/2}\hat{\sigma}] \cup [\theta_0 + z_{\alpha/2}\hat{\sigma}, +\infty)$;
2. $P-$value: reject H_0 if $P-$value $\hat{p} = 1 - \Big(\Phi\big(|\frac{\widehat{\theta}-\theta_0}{\widehat{\sigma}}|\big) - \Phi\big(-|\frac{\widehat{\theta}-\theta_0}{\widehat{\sigma}}|\big)\Big) < \alpha$, i.e., $|\frac{\widehat{\theta}-\theta_0}{\widehat{\sigma}}| > z_{\alpha/2}$;
3. Confidence Interval: if $\widehat{\theta} \notin (\theta_0 - z_{\alpha/2}\hat{\sigma}, \theta_0 + z_{\alpha/2}\hat{\sigma})$.

Similarly, for one-sided test $H_0 : \theta = \theta_0$ vs. $H_1 : \theta > \theta_0$ (or $H_1 : \theta < \theta_0$), the three equivalent ways translate to

1. Rejection region: $R(\alpha) = [\theta_0 + z_\alpha\hat{\sigma}, +\infty)$; (or $R(\alpha) = (-\infty, \theta_0 - z_\alpha\hat{\sigma})$);
2. $P-$value: reject H_0 if $P-$value $\hat{p} = 1 - \Phi\big(\frac{\widehat{\theta}-\theta_0}{\widehat{\sigma}}\big) < \alpha$, i.e., $\frac{\widehat{\theta}-\theta_0}{\widehat{\sigma}} > z_\alpha$; (or $\hat{p} = \Phi\big(\frac{\widehat{\theta}-\theta_0}{\widehat{\sigma}}\big) < \alpha$);
3. Confidence Interval: if $\widehat{\theta} \notin (-\infty, \theta_0 + z_\alpha\hat{\sigma})$. (or $\widehat{\theta} \notin (\theta_0 - z_\alpha\hat{\sigma}, +\infty)$)

References

Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman (2018). *Probability and statistical inference.*